

FPGA Development for Radar, Radio-Astronomy and Communications

THE
RADAR
MASTERS COURSE



Dept. Electrical Engineering, University of Cape Town
Private Bag, Rondebosch, 7701, South Africa
<http://www.rmsg.uct.ac.za>



Presented by Dr John-Philip Taylor
Convened by Dr Stephen Paine

Day 1 – 3 August 2026

Outline

Introduction

FPGA Internals

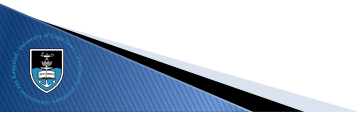
The Development Kit

Development Cycle

Verilog Basics

Verilog Processes

IP Library



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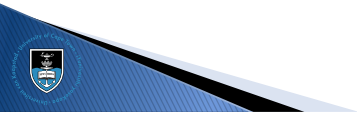
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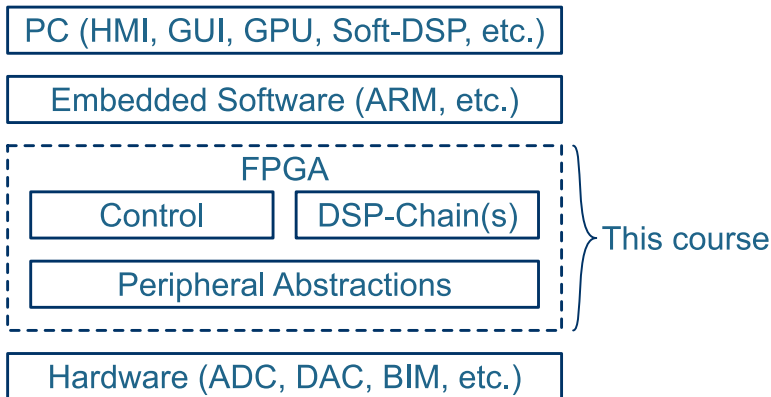
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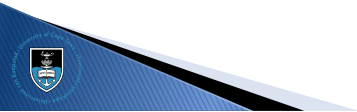


Course Overview



Course Overview

- ▶ Overall (Morning)
 - ▶ This course specifically targets low-level development
 - ▶ Minimal exposure to vendor-provided IP and related workflows
 - ▶ No high-level synthesis (HLS)
 - ▶ No soft-core or embedded processors / etc.
- ▶ Practicals (Afternoon)
 - ▶ Practical descriptions are available on the [course Git repository](#).
- ▶ Project and Report (After the course)
 - ▶ Each participant needs to implement a project and hand in a report for evaluation.
- ▶ **Never be shy to ask questions**



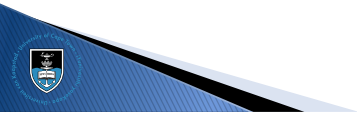
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- ▶ Practicals (Afternoon)
 - ▶ Practical descriptions are available on the [course Git repository](#).
 - ▶ The practicals are challenging to finish during the scheduled tutorial sessions.
 - ▶ The scheduled tutorial sessions are informal – feel free to structure your time as you see fit, or even do the practicals at home.
 - ▶ Fork the repository and commit your work. You can mark the project as private, but add the presenter as a collaborator in that case.
 - ▶ Primarily Verilog, but feel free to implement your practicals in VHDL (the presenter is well-versed in both).
- ▶ Project and Report (After the course)
 - ▶ Each participant needs to implement a project and hand in a report for evaluation.
- ▶ **Never be shy to ask questions**



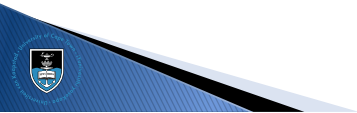
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- ▶ Project and Report (After the course)
 - ▶ Each participant needs to implement a project and hand in a report for evaluation.
 - ▶ Each participant needs to implement something different – some ideas will be presented at the end of the course.
 - ▶ Submit your report on [Amathuba](#).
 - ▶ The report should contain a link to your source code repository, which could be the fork you used for the practicals.
- ▶ Never be shy to ask questions



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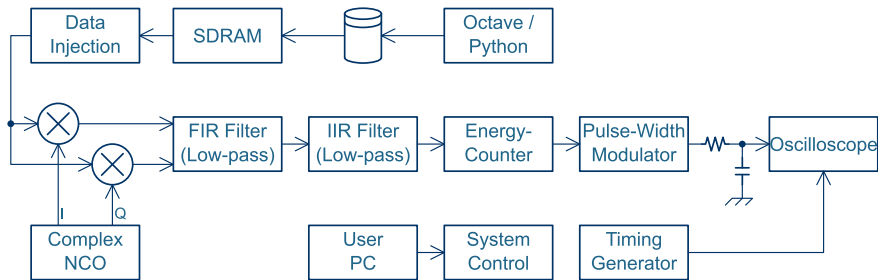


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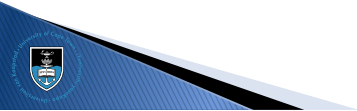
- ▶ Understand the underlying physical architecture of FPGAs
- ▶ Understand the concept of timing constraints, clock domains and other timing-related issues
- ▶ Use the FPGA tool-set, including JTAG debugging and the general Verilog-based compilation process
- ▶ Design FPGA firmware systems on a high level
- ▶ Design RTL representations of finite state machines and pipelines
- ▶ Implement FPGA firmware systems
- ▶ Debug an FPGA firmware implementation
- ▶ Analyse timing closure issues and solve the problem such that the final design meets all timing requirements.



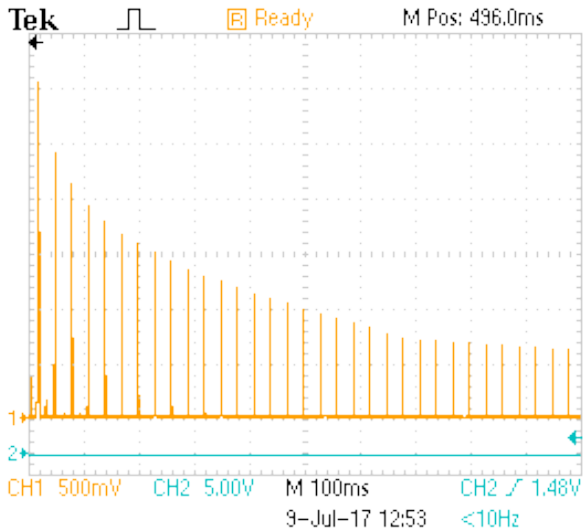
Practicals



- ▶ JTAG interfacing to / from the computer
- ▶ 8-bit, 100 MSps data injection
- ▶ 1024-point FIR-filter with a $128\times$ subsampling rate

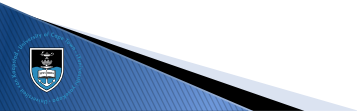


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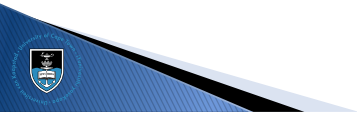
The Audience

- ▶ Programming experience?
 - ▶ C / C++ / C#?
 - ▶ Embedded? PC?
 - ▶ GPU? OpenGL? OpenCL? DirectX? CUDA?
- ▶ FPGA Experience?
 - ▶ VHDL? Verilog? PyHDL? Migen? Graphical tools?
 - ▶ Synthesis? Simulation only?
 - ▶ What context / application?



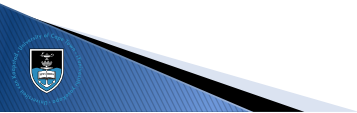
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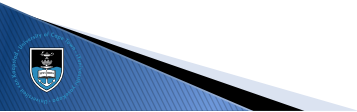
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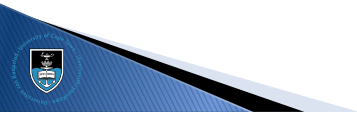
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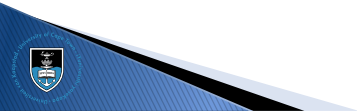
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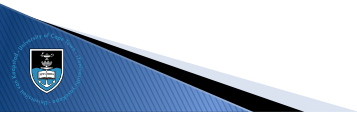
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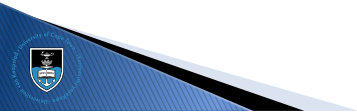
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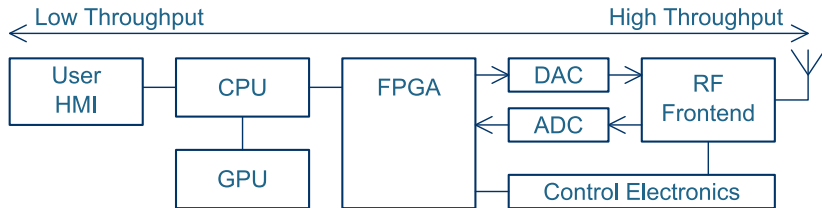


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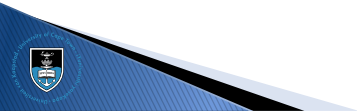
Software-Defined Radio



- ▶ In high-performance computing, the CPU, GPU and FPGA work together
- ▶ Each processing element is used for its strength

Processing Platforms

- ▶ CPU:
 - ▶ General-purpose; Easily modified; Short development cycle
 - ▶ Extensive history $\Rightarrow$ Rich set of matured libraries and APIs
 - ▶ Does not handle parallel algorithms particularly well
- ▶ GPU:
- ▶ FPGA:



Processing Platforms

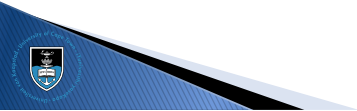
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▶ GPU:

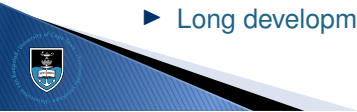
- ▶ Agile: can be reprogrammed in real-time (kernels are referenced by a handle, or memory address)
- ▶ Extremely good at coarse-grained parallel algorithms (little to none inter-worker communication: matrix multiplication; FFT; FIR filters; etc.)
- ▶ Medium development cycle – debugging and optimising is a challenging process
- ▶ Memory bandwidth problems (the bottle-neck is generally in data transfer, but less so in modern GPUs with advanced parallel copy-engines)

▶ FPGA:

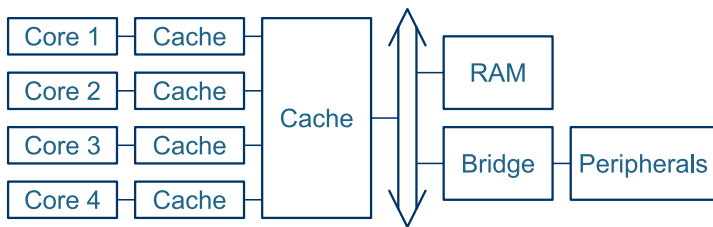


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 - ▶ Medium development cycle
- ▶ FPGA:
 - ▶ Not agile: takes a few seconds to switch functionality
 - ▶ Highly flexible, but with a relatively slow clock
 - ▶ Provides ASIC functionality at a small fraction of the cost
 - ▶ Excellent at fine-grained and time-critical problems
 - ▶ Long development cycle; challenging to debug and optimise

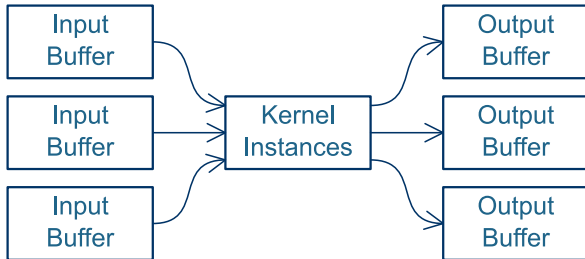


Programming Model – CPU



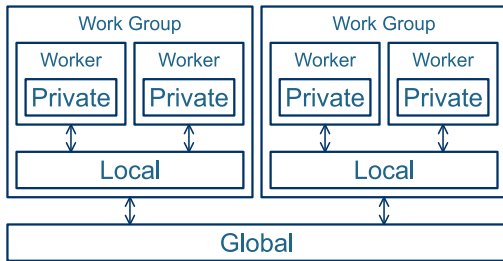
- ▶ Fetch-decode-execute cycle – serialised execution
- ▶ The RAM is divided into program, stack and heap areas
- ▶ All CPU cores share the same RAM, but is cache-assisted to reduce contention

Programming Model – GPU



- ▶ Client-server interface with the CPU
- ▶ Runs a pipeline of kernels, in SIMD operation

Programming Model – GPU

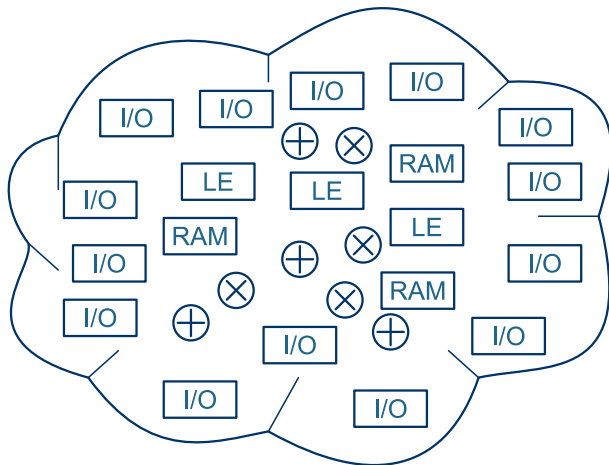


- ▶ The hardware is organised into groups of processors
- ▶ Within a group, execution is in lock-step
- ▶ Execution within one group is independent of other groups
- ▶ Three levels of memory



Programming Model – FPGA

- Any architecture you like...



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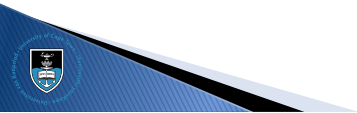
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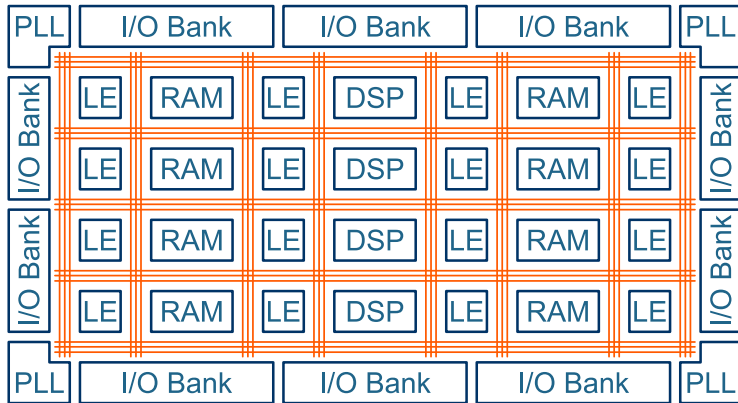
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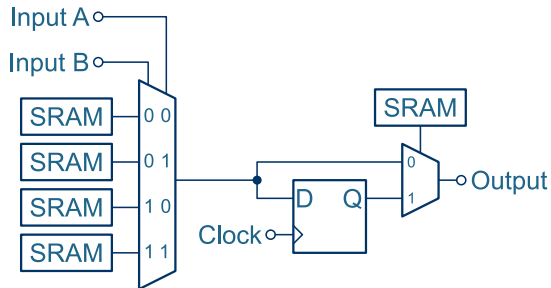
IP Library



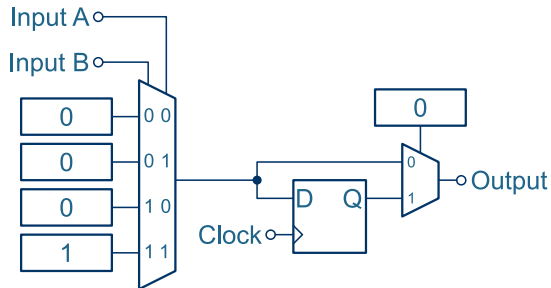
FPGA Architecture Overview



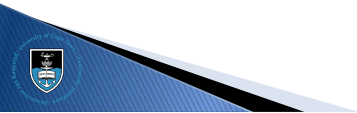
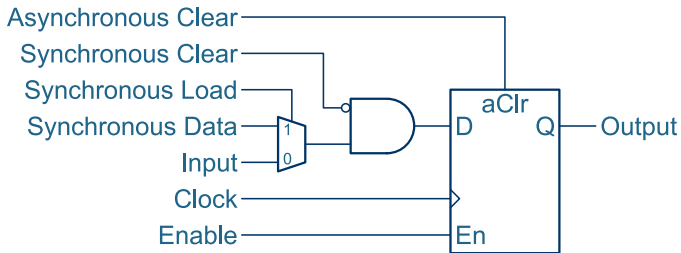
Logic Elements



Logic Elements

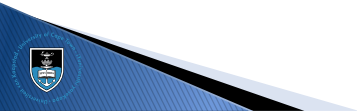


Register Detail

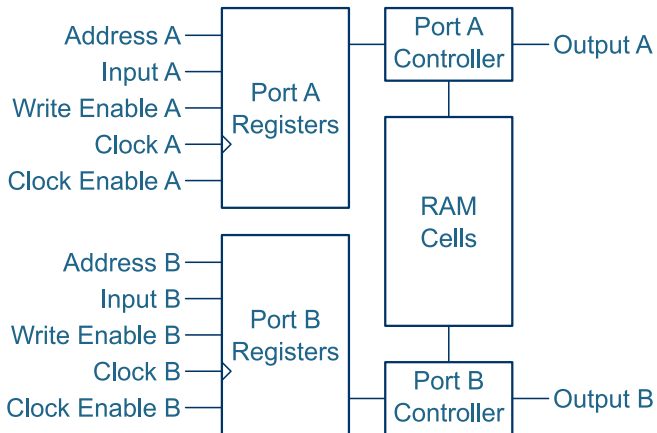


Register Detail

A-Clr	En	S-Clr	S-Ld	S-Dat	In	Clk	Output
1	×	×	×	×	×	×	0
0	1	1	×	×	×	↑	0
0	1	0	1	0	×	↑	0
0	1	0	1	1	×	↑	1
0	1	0	0	×	0	↑	0
0	1	0	0	×	1	↑	1
0	0	×	×	×	×	×	no-change



Internal RAM Blocks



Internal RAM Blocks

- ▶ MAX-10 M9K ports can be configured as:

8192 × 1

4096 × 2

2048 × 4

1024 × 8

1024 × 9

512 × 16

512 × 18

256 × 32

256 × 36

- ▶ The two ports can have different configurations
- ▶ The two ports can have independent clocks



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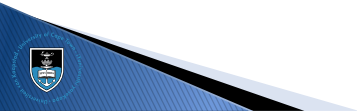
- Internal RAM can be initialised by means of a memory initialisation file (.mif)

```
-- Header Comments
```

```
WIDTH=8;  
DEPTH=4096;
```

```
ADDRESS_RADIX=HEX;  
DATA_RADIX=HEX;
```

```
CONTENT BEGIN  
    [000..011] : 00;  
    012        : 7E;  
    013        : 81;  
    ...  
    [FFD..FFE] : FF;  
    FFF        : 00;  
END;
```

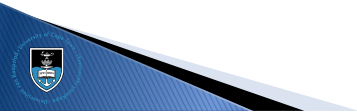


Internal RAM Blocks

- ▶ Internal RAM can be initialised by means of a memory initialisation file (.mif)
- ▶ Xilinx (AMD) uses COE files

```

MEMORY_INITIALIZATION_RADIX=16;
MEMORY_INITIALIZATION_VECTOR=
00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00,
00, 00, 7E, 81, A5, 81, 81, BD, 99, 81, 81, 7E, 00, 00, 00, 00,
00, 00, 7E, FF, DB, FF, FF, C3, E7, FF, FF, 7E, 00, 00, 00, 00,
...
00, 70, D8, 30, 60, C8, F8, 00, 00, 00, 00, 00, 00, 00, 00, 00,
00, 00, 00, 00, 7C, 7C, 7C, 7C, 7C, 7C, 7C, 00, 00, 00, 00, 00,
00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, 00, FF, FF, 00;
  
```



Internal RAM Blocks

- ▶ Internal RAM can be initialised by means of a memory initialisation file (.mif)
- ▶ Xilinx (AMD) uses COE files
- ▶ Lattice uses .mem files)

```
// Header Comments
#Format=Hex
#Depth=4096
#Width=8
#AddrRadix=3
#DataRadix=3
#Data
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 7E 81 A5 81 81 BD 99 81 81 7E 00 00 00 00
00 00 7E FF DB FF FF C3 E7 FF FF 7E 00 00 00 00
...
00 70 D8 30 60 C8 F8 00 00 00 00 00 00 00 00 00
00 00 00 00 7C 7C 7C 7C 7C 7C 7C 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF FF 00
```



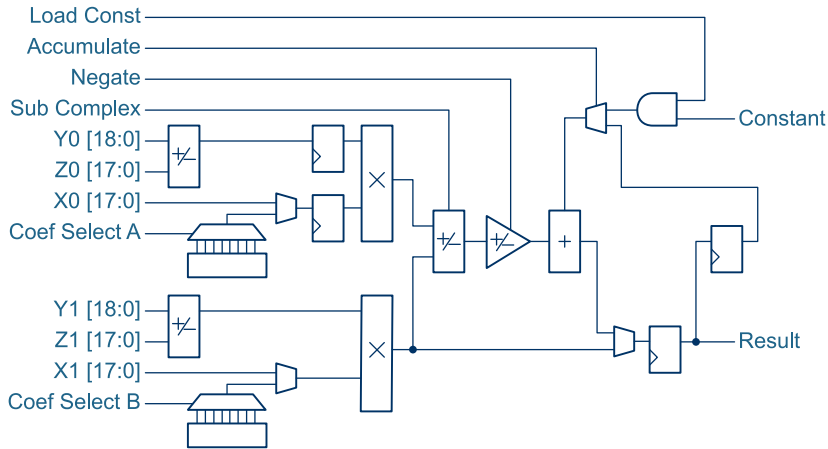
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```
// Header Comments
#Format=AddrHex
#Depth=4096
#Width=8
#AddrRadix=3
#DataRadix=3
#Data
012:7E 81 A5 81 81 BD 99 81 81 7E
022:7E FF DB FF FF C3 E7 FF FF 7E
034:6C FE FE FE FE 7C 38 10
...
FD1:70 D8 30 60 C8 F8
FE4:7C 7C 7C 7C 7C 7C 7C
FFD:FF FF
```



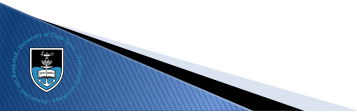
DSP Blocks (Altera / Intel)



DSP Detail (Altera / Intel)

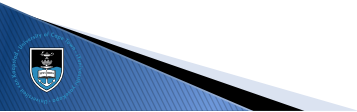
Each Cyclone V DSP block can be configured as:

- ▶ Three 9×9 -bit multipliers
- ▶ Two 18×18 -bit multipliers (unsigned)
- ▶ Two 18×19 -bit multipliers (signed)
- ▶ One 18×25 -bit multiplier
- ▶ One 20×24 -bit multiplier
- ▶ One 27×27 -bit multiplier
- ▶ One 18×19 -bit multiply-accumulate
- ▶ One 18×18 -bit multiply-accumulate with adder
- ▶ Half of a 18×19 -bit complex multiplier



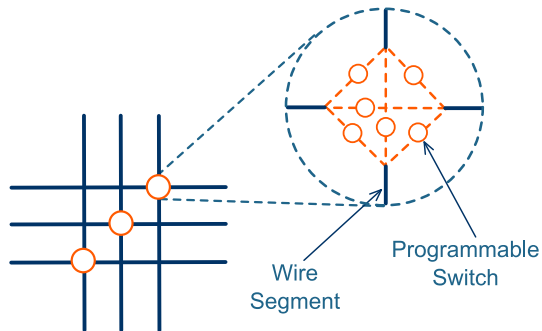
MAX-10 Multipliers

- ▶ The MAX-10 does not have DSP blocks – it only has embedded multipliers (no accumulators or adders as part of the block)
- ▶ Each multiplier block can be configured as two 9×9 -bit multipliers or one 18×18 -bit multiplier
- ▶ Each input can be configured as signed or unsigned
- ▶ The inputs and outputs can optionally be registered inside the multiplier block, which improves timing



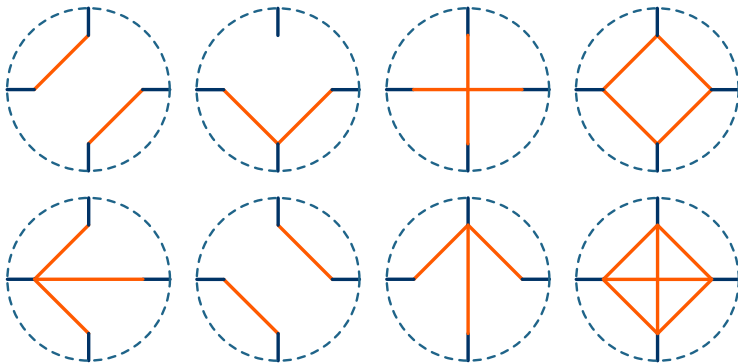
Interconnect

- ▶ Each interconnect crossing contains 6 switches
- ▶ These switches can be configured in various ways



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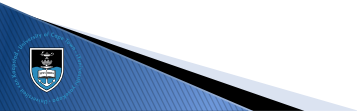
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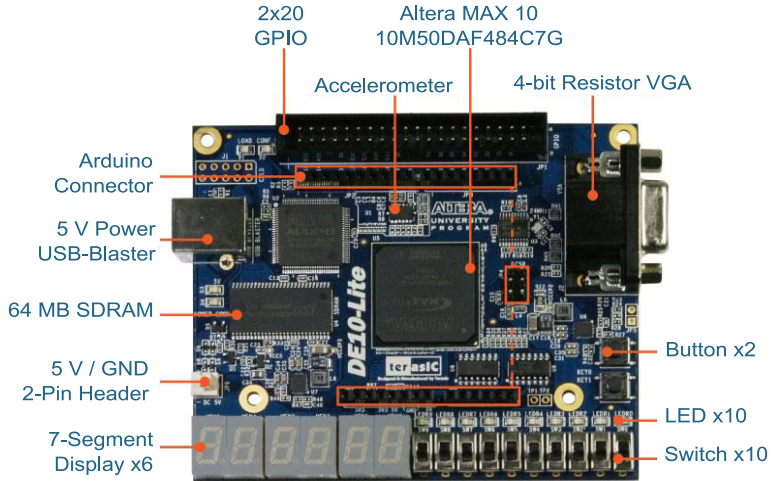
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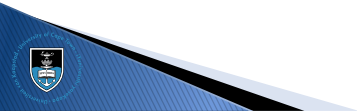


DE10 Lite Overview



FPGA Details

- ▶ MAX-10 series (10M50DAF484C7G)
- ▶ 2 ADCs (each 12-bit, 1 MSps shared over 9 channels)
- ▶ 49 760 logic elements
- ▶ 182 M9K memory blocks
- ▶ 736 kiB user flash memory
- ▶ 144 multiplier blocks
- ▶ 4 phase-locked loop blocks
- ▶ 360 I/O Pins



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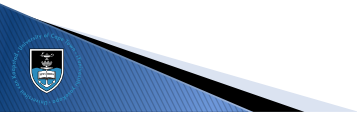
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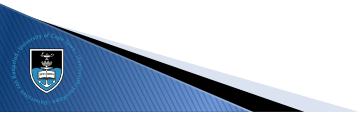
Verilog Basics

Verilog Processes

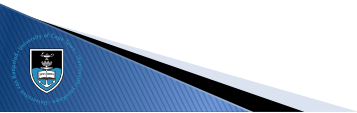
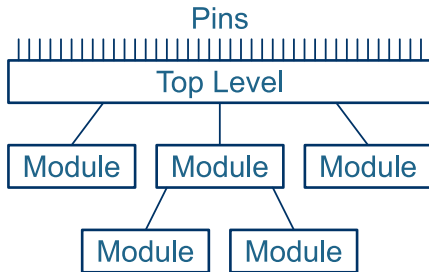
IP Library



Development Cycle



Design Entry – Modular Design



Design Entry – HDL

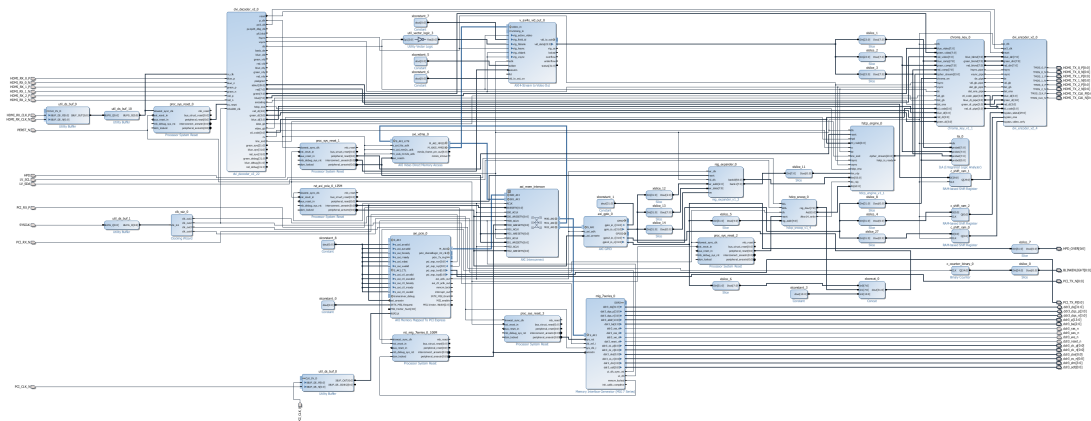
- Verilog / VHDL / PyHDL / Migen / etc.

```
module Top_Level(input ipClk, input ipButton, output opLED);  
  wire Debounced_Button;  
  
  Debouncer Debouncer_Inst(  
    ipClk, ipButton, Debounced_Button  
  );  
  
  LED_Driver LED_Driver_Inst(  
    ipClk, Debounced_Button, opLED  
  );  
endmodule
```

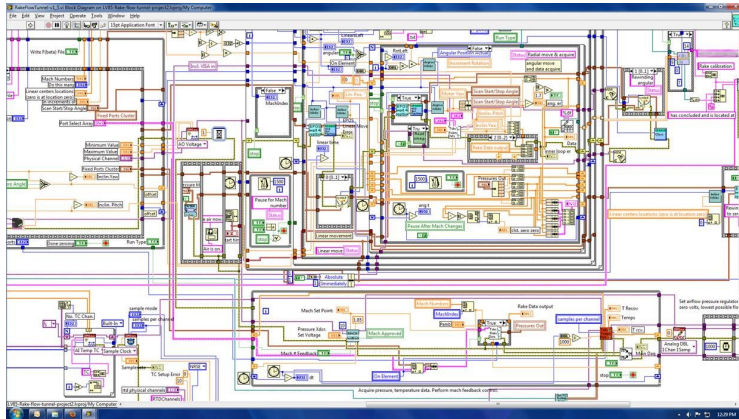


Design Entry – Schematic





Design Entry – Schematic





Design Entry – Platform Designer

System Contents Address Map Interconnect Requirements						
System: QSys Path: master						
Use	Connections	Name	Description	Export	Clock	Base
☒		clk_0	Clock Source			
		clk_in	Clock Input	clk	exported	
		clk_in_reset	Reset Input	reset		
		clk	Clock Output	Double-click to export	clk_0	
		clk_reset	Reset Output	Double-click to export		
☒		JTAG	JTAG to Avalon Master Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		clk_reset	Reset Input	Double-click to export		
		master	Avalon Memory Mapped Master	Double-click to export	[clk]	
		master_reset	Reset Output	Double-click to export		
☒		master	Avalon-MM Pipeline Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		reset	Reset Input	Double-click to export	[clk]	
		s0	Avalon Memory Mapped Slave	master	[clk]	
		m0	Avalon Memory Mapped Master	Double-click to export	[clk]	
☒		SDRAM	Avalon-MM Pipeline Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		reset	Reset Input	Double-click to export	[clk]	
		s0	Avalon Memory Mapped Slave	Double-click to export	[clk]	
		m0	Avalon Memory Mapped Master	sdram	[clk]	# 0x0000_0000
☒		Registers	Avalon-MM Pipeline Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		reset	Reset Input	Double-click to export	[clk]	
		s0	Avalon Memory Mapped Slave	Double-click to export	[clk]	
		m0	Avalon Memory Mapped Master	registers	[clk]	# 0x0400_0000



Design Entry – HLS

```

void paralleltest(
    bool _doWrite, int _writeAddr, int _writeData,
    bool _doRead, int _readAddr, int* _readData
){
    #pragma HLS INTERFACE ap_ctrl_none port=return
    #pragma HLS PIPELINE II=1
    #pragma HLS DEPENDENCE variable=buffer inter WAR false
    #pragma HLS RESOURCE    variable=buffer core=RAM_2P_BRAM

    static const int BufferSize = 1024;
    static          int buffer[BufferSize];

    if (_doWrite){buffer[_writeAddr % BufferSize] = _writeData;}
    if (_doRead) {*_readData = buffer[_readAddr % BufferSize];}
}

```



Analysis and Synthesis

- ▶ Analyse code and perform optimisations (trim dead paths, check connectivity, etc.)
- ▶ Synthesise logic tables from expressions
- ▶ Match the logic tables and data flow graphs to the target hardware architecture
- ▶ Synthesise connection graphs

Note: Verilog automatically generates a 1-bit wire for any net that is used without definition, which often happens on typo's and spelling errors. Consult the synthesis warnings for “not declared” nets.



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Fitter

- ▶ The fitter performs a place and route function (similar to a PCB auto-router)
- ▶ This is a computationally-intensive process, so be patient
- ▶ Uses design constraints:
 - ▶ Clock rate
 - ▶ Combinational logic time delay
 - ▶ Multi-cycle timing requirements
 - ▶ False-path specifications
 - ▶ Routing delay
 - ▶ External timing requirements
 - ▶ User-defined placement
 - ▶ etc.



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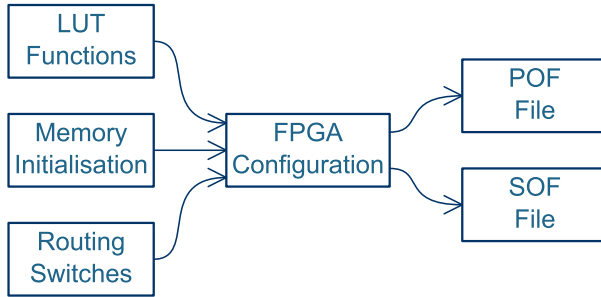


Fitter

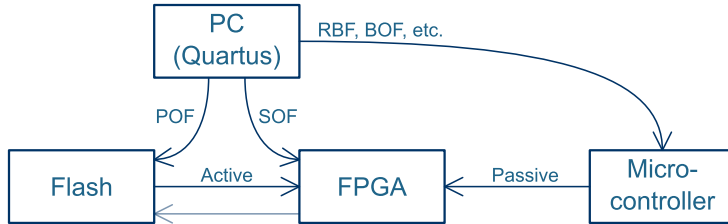
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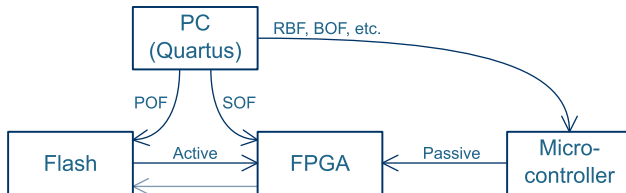
Assembler



Programming Architectures



Programming Architectures

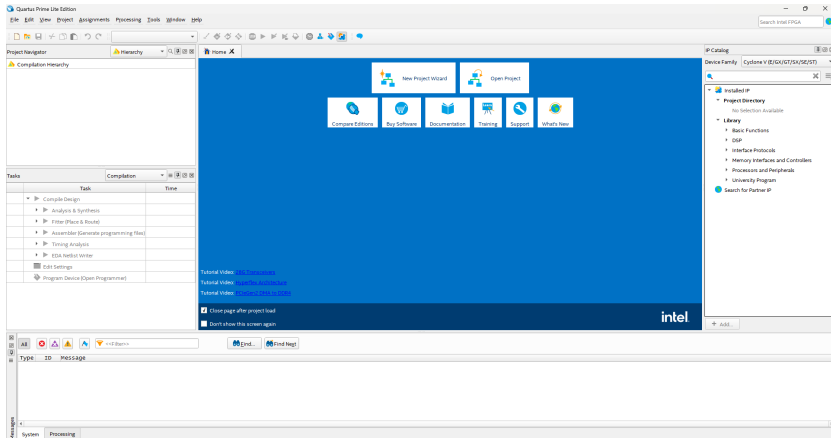


- ▶ SOF $\Rightarrow$ SRAM Object File
- ▶ POF $\Rightarrow$ Programmer Object File
- ▶ RBF $\Rightarrow$ Raw Binary File
- ▶ BOF $\Rightarrow$ Borph Object File

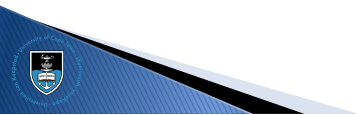


Altera Quartus

Practical 01 – Quartus introduces the Altera Quartus IDE.



Coffee Break...



Outline

Introduction

FPGA Internals

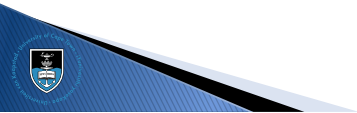
The Development Kit

Development Cycle

Verilog Basics

Verilog Processes

IP Library



Module Definition

```
/* Use comments to describe the function...
   Multiple lines are possible */

module MyModule(
    input      ipClk,    // Try to be consistent
    input      ipReset, // on port placement

    input  [ 7:0]ipInput, // Note that Verilog is
    output [ 9:0]opOutput, // case sensitive
    inout  [12:0]bpBidirectional
);

// The module body goes here...

endmodule
```

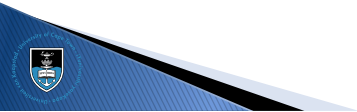


Wires

- ▶ Wires are internal connections
- ▶ See them as wires on a bread-board...

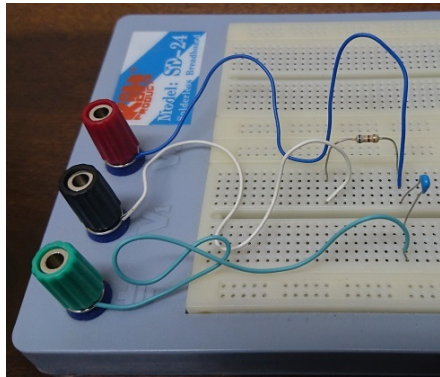
```
wire      A;          // A single-bit wire
wire [7:0] B;          // An 8-bit wire
wire [7:0] C[5:0];     // A 6-element array of 8-bit wires

wire [7:0] X = B + C[3]; // Wire definition and
                        // assignment in one
```



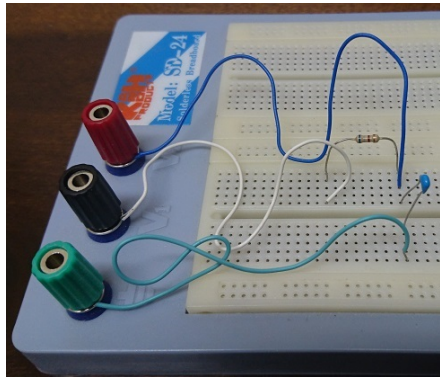
Wires

```
module PWM_Filter(  
    input  V1,  
    output V2,  
    input  GND  
);  
    wire Blue;  
    wire White = V1;  
    wire Cyan = GND;  
  
    Resistor #( 680) R1(White, Blue);  
    Capacitor #(10000) C1(Blue , Cyan);  
    assign V2 = Blue;  
endmodule
```



Wires

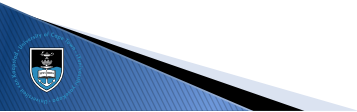
```
module PWM_Filter(  
    input  V1,  
    output V2,  
    input  GND  
);  
    Resistor #( 680) R1(V1, V2 );  
    Capacitor #(10000) C1(V2, GND);  
endmodule
```



Operators – Reduction

```
wire [7:0] X, Y, Z; // Defines 3x 8-bit wires
wire      A, B, C; // Defines 3x 1-bit wires

assign A = &X; // AND-reduce X and assign to A
assign B = |Y; // OR-reduce Y and assign to B
assign C = ^Z; // XOR-reduce Z and assign to C
assign B = !Y; // Logical NOT: equivalent to B = ~|Y
```



Operators – Bitwise

```
wire [7:0] X, Y, Z; // Defines 3x 8-bit wires

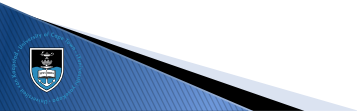
assign Z = -X;      // 2's complement X and assign to Z
assign Z = ~Y;      // Bitwise NOT Y and assign to Z
assign Z = X | Y;    // Bitwise OR X with Y and assign to Z
assign Z = X & Y;    // Bitwise AND X with Y and assign to Z
assign Z = X ^ Y;    // Bitwise XOR X with Y and assign to Z
```



Operators – Concatenation

```
wire [7:0] A;  
wire [3:0] B, C;
```

```
assign A = {B, C}; // Concatenates B--C and assign to A  
assign {B, C} = A; // Assign A to the concatenation B--C  
assign A = {2{B}}; // Replicate B 2 times and assign to A
```

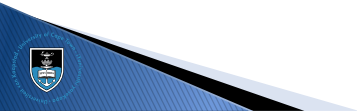


Operators – Arithmetic

```
wire [ 7:0] A, B, C;  
wire [15:0] X;
```

```
assign A = B + C; // Add C to B and assign to A  
assign A = B - C; // Subtract C from B and assign to A  
assign X = B * C; // Multiply B by C and assign to X
```

```
assign A = B << 5; // Left-shift B and assign to A  
assign A = B >> 6; // Right-shift B and assign to A
```



Operators – Logical

```
wire [ 7:0]A, B, C;  
wire      X;
```

```
assign X = A > B; // A greater than B?      Result to X
```

```
assign X = A < B; // A less than B?          Result to X
```

```
assign X = A >= B; // A greater or equal to B? Result to X
```

```
assign X = A <= B; // A less or equal to B?   Result to X
```

```
assign X = A == B; // A equal to B?          Result to X
```

```
assign X = A != B; // A not equal to B?       Result to X
```

```
assign X = A && B; // Equivalent to X = (|A) & (|B)
```

```
assign X = A || B; // Equivalent to X = (|A) | (|B)
```

```
assign C = X ? A : B // If X is 1, assign A to C,  
                    // otherwise assign B to C
```

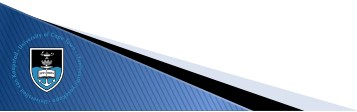


Operators – Signed Operations

- ▶ All operations are unsigned, unless specified otherwise
- ▶ The following always yield unsigned results:
 - ▶ Any operation with at least one unsigned operand
 - ▶ A literal without the 's' modifier
 - ▶ Bit-select (eg. A[5])
 - ▶ Part-select (eg. A[5:3])
 - ▶ Concatenations

```
wire          [ 7:0]A; // Unsigned vector  
wire signed [ 7:0]B; // Signed vector  
wire signed [15:0]X; // Signed vector
```

```
assign X = $signed(A) * B; // A must be cast
```



Operators – Signed Operations

- ▶ In general, most operations should be done using standard unsigned arithmetic
- ▶ In the rare cases where the sign makes a difference (multiplication and relational statements), cast explicitly

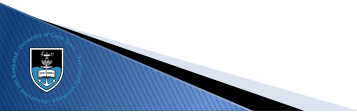
```
wire [ 7:0]A;
```

```
wire [ 7:0]B;
```

```
wire [15:0]X;
```

```
assign X = $signed(A) * $signed(B);
```

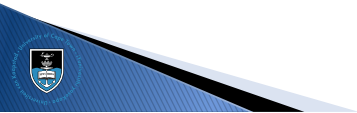
```
PWM PWM1(Clk, {~X[15], X[14:0]}, PWM_Output);
```



Vector Lengths

Note: Verilog does not enforce vector length matching:

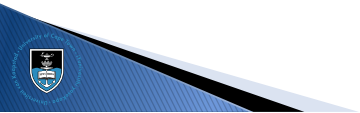
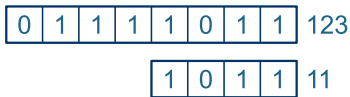
- ▶ If the left-hand-side is longer than the right-hand-side, the most-significant side is padded with zeros
- ▶ If the left-hand-side is shorter than the right-hand-side, the most-significant side is truncated
- ▶ **This is done without an obvious warning!**
- ▶ Consult the compile warnings to check for unintended size-mismatches



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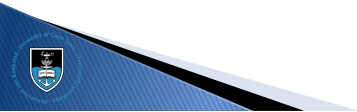
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- ▶ **This is done without an obvious warning!**
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Literals

```
// All underscores "_" are ignored
// (except in Lattice: leading "_" gives an error)
5'b_1_0110 // 5-bit binary
11'h_5_CE // 11-bit hexadecimal
13'o_1_7642 // 13-bit octal
17'd_123_456 // 17-bit decimal

"H" // 8-bit ASCII constant

// Left bits (most significant) are padded with zeros,
// unless the most significant specified is 'Z' or 'X'
78'bZ // 78-bit high-impedance

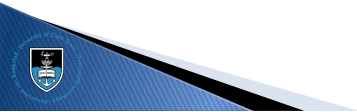
// Use the 's' specifier for "signed" literals
-8'sd125 // 2's complement of the positive
        // signed decimal constant 125
```



Module Instances

```
module My_Submodule(input Clk, input A, input B, output X);  
    // Module body...  
endmodule
```

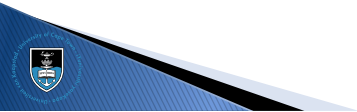
```
module Top_Level(  
    input  Clock_50MHz,  
    input  Input1, Input2,  
    output Output  
);  
  
// Positional port mapping:  
My_Submodule Instance_1(  
    Clock_50MHz,  
    Input1, Input2, Output  
);
```



Module Instances

```
module My_Submodule(input Clk, input A, input B, output X);  
    // Module body...  
endmodule
```

```
module Top_Level(  
    input  Clock_50MHz,  
    input  Input1, Input2,  
    output Output  
);  
  
// Named port mapping:  
My_Submodule Instance_2(  
    .X(Output), .A(Input1), .B(Input2),  
    .Clk(Clock_50MHz)  
);
```



Outline

Introduction

FPGA Internals

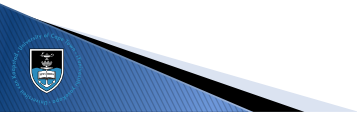
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Blocking Statements

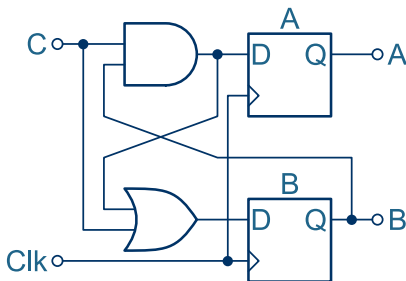
```
reg A, B;
```

```
always @ (posedge ipClk) begin
```

A = C & B;

```
B = A | C;
```

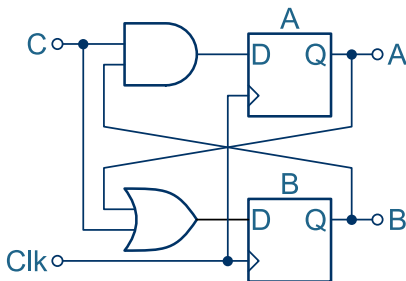
end



- ▶ Statements are evaluated in order, like a computer program
- ▶ Often results in unintentionally long combinational chains
- ▶ Note that all registers still change state on the clock edge

Non-Blocking Statements

```
reg A, B;  
  
always @(posedge ipClk) begin  
    A <= C & B;  
    B <= A | C;  
end
```

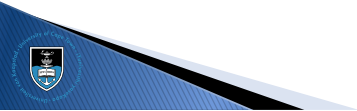


- ▶ All right-hand-side expressions are evaluated in parallel and then assigned to the left-hand-side on the clock edge
- ▶ The order of statements makes no difference to the functionality

Mixed Mode

**Never mix blocking and non-blocking statements
in the same always block**

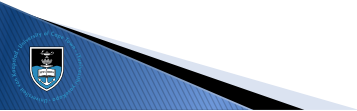
Except inside test-benches, where it is sometimes useful...



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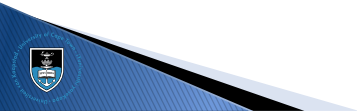


Algorithmic Combinational Blocks

- Use blocking assignments: allows algorithmic descriptions

```
wire [7:0]Byte; // Assigned outside always block => wire
reg [2:0]Count; // Assigned inside always block => reg
integer n;      // Non-synthesisable type
                  // (compilation-time only)

always @(*) begin
    Count = 0;
    for(n = 0; n < 8; n = n+1) begin
        Count = Count + Byte[n];
    end
end
```



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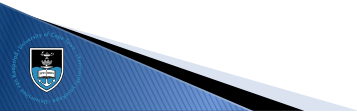
always @(*) begin
    Count = 0;
    for(n = 0; n < 8; n = n+1) begin
        Count = Count + Byte[n];
    end
end
```

- If not explicitly assigned a new value,
the previous value is “remembered” in an “inferred latch” – to be avoided



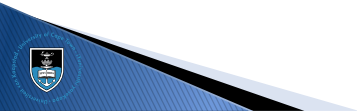
Look-up Tables

```
always @(*) begin
  case (BCD)
    4'h0:    SevenSegment = 7'b0111111;
    4'h1:    SevenSegment = 7'b0000110;
    4'h2:    SevenSegment = 7'b1011011;
    4'h3:    SevenSegment = 7'b1001111;
    4'h4:    SevenSegment = 7'b1100110;
    4'h5:    SevenSegment = 7'b1101101;
    4'h6:    SevenSegment = 7'b1111101;
    4'h7:    SevenSegment = 7'b0000111;
    4'h8:    SevenSegment = 7'b1111111;
    4'h9:    SevenSegment = 7'b1101111;
    default:; // This is bad: infers a latch
  endcase
end
```



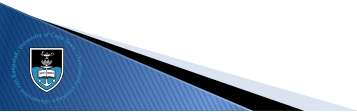
Look-up Tables

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always @(*) begin
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    4'h7:    SevenSegment = 7'b0000111;
    4'h8:    SevenSegment = 7'b1111111;
    4'h9:    SevenSegment = 7'b1101111;
    default: SevenSegment = 0; // This is acceptable
  endcase
end
```



Look-up Tables

```
always @(*) begin
  case (BCD)
    4'h0:    SevenSegment = 7'b0111111;
    4'h1:    SevenSegment = 7'b0000110;
    4'h2:    SevenSegment = 7'b1011011;
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    4'h7:    SevenSegment = 7'b0000111;
    4'h8:    SevenSegment = 7'b1111111;
    4'h9:    SevenSegment = 7'b1101111;
    default: SevenSegment = 7'bXXXXXXX; // This is better
  endcase
end
```



Sparse Case Statements

```
always @(*) begin
  case (Address) // 8-bit address, 16-bit data
    8'h00: Data = FirmwareVersion;
    8'h01: Data = BuildDate;
    8'h02: Data = BuildTime;

    8'h10: Data = { 6'd0, LED      };
    8'h11: Data = { 6'd0, Switches};
    8'h12: Data = {14'd0, Buttons };

    8'h20: Data = Accelerometer_X;
    8'h21: Data = Accelerometer_Y;
    8'h22: Data = Accelerometer_Z;

    default: Data = 0; // This is OK
  endcase
end
```



Sparse Case Statements

```
always @(*) begin
  case (Address) // 8-bit address, 16-bit data
    8'h00: Data = FirmwareVersion;
    8'h01: Data = BuildDate;
    8'h02: Data = BuildTime;

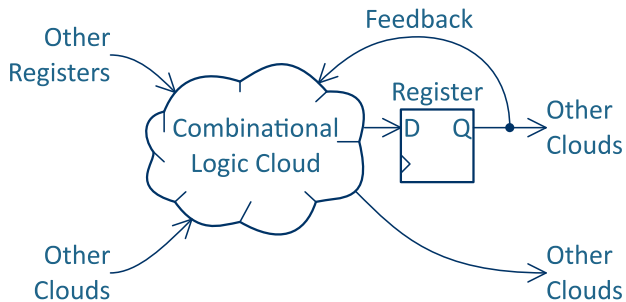
    8'h10: Data = { 6'd0, LED      };
    8'h11: Data = { 6'd0, Switches};
    8'h12: Data = {14'd0, Buttons };

    8'h20: Data = Accelerometer_X;
    8'h21: Data = Accelerometer_Y;
    8'h22: Data = Accelerometer_Z;

    default: Data = 16'hXXXX; // This is better
  endcase
end
```



Register Transfer Logic



Register Transfer Logic

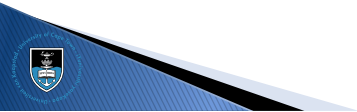
- Use non-blocking assignments: easier to relate all calculations to the clock edge

```
reg Reset;

always @(posedge ipClk) begin
    Reset <= ipReset; // Localise the reset

    if(Reset) begin
        // Reset stuff here

    end else if(ipEnabled) begin
        // RTL code goes here
    end
end
```



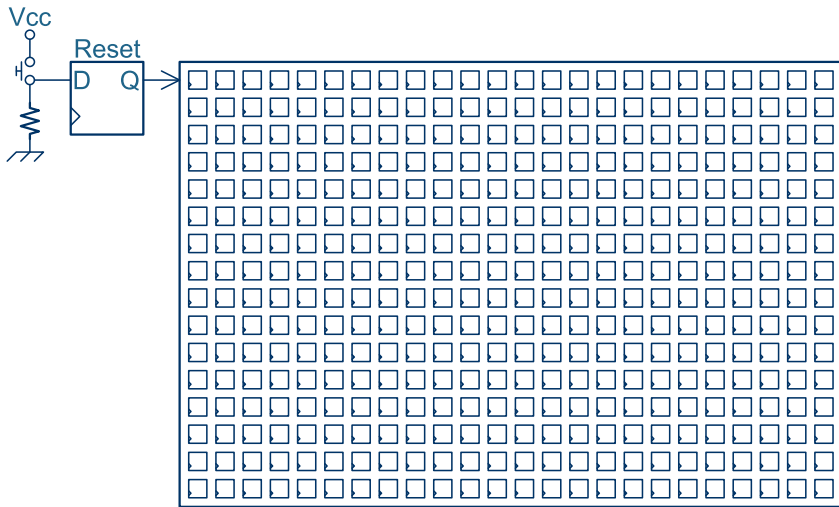
Register Transfer Logic

- If not explicitly assigned a new value, the previous value is “remembered” in the register – very useful

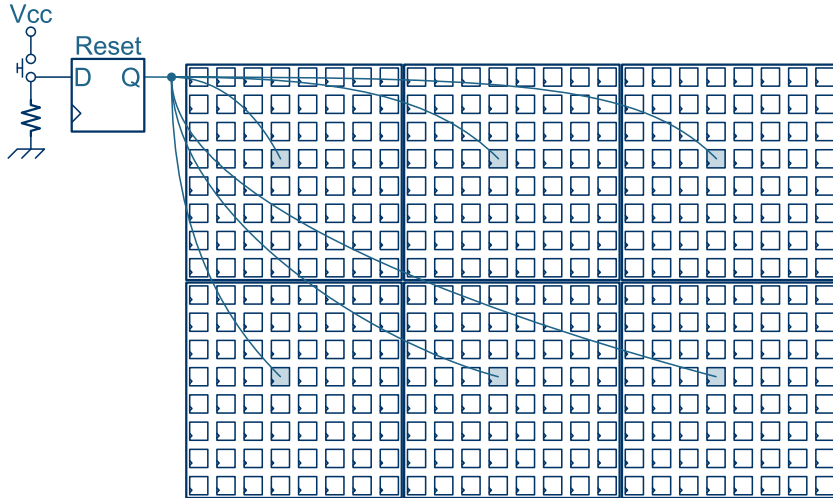
```
reg      Reset;  
reg [11:0] Count;  
  
always @(posedge ipClk) begin  
    Reset <= ipReset; // Localise the reset  
  
    if(Reset) begin  
        Count <= 0;  
  
    end else if(ipEnabled) begin  
        Count <= Count + 1'b1;  
    end  
end
```



Synchronous and Local Resets



Synchronous and Local Resets



Outline

Introduction

FPGA Internals

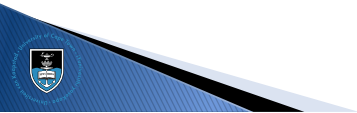
The Development Kit

Development Cycle

Verilog Basics

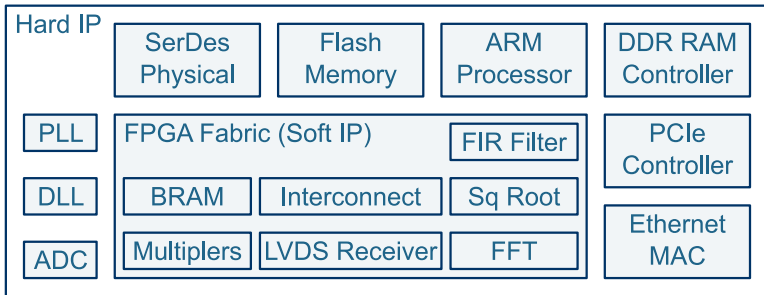
Verilog Processes

IP Library



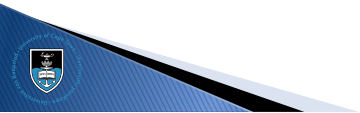
IP Library

- ▶ A combination of soft and hard IP
- ▶ Collectively known as “Megafunctions” or “IP Cores”



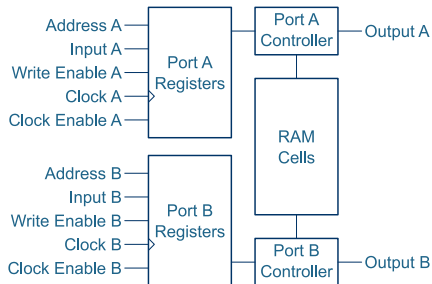
Wizard

- ▶ Generally easier to use the IP Catalogue (or Platform Designer) wizard to generate wrapper modules
- ▶ Or one can instantiate the built-in modules directly



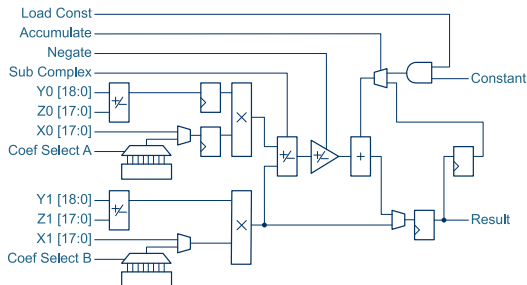
Embedded Components

- ▶ RAM / ROM
- ▶ DSP blocks
- ▶ PLL / DLL blocks
- ▶ Processors / SoC (ARM) with bus infrastructure
- ▶ Interfaces (DDR Memory / PCIe / SerDes (JESD204) / etc.)



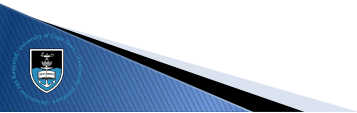
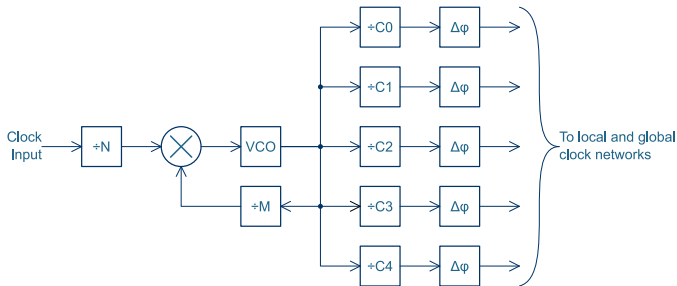
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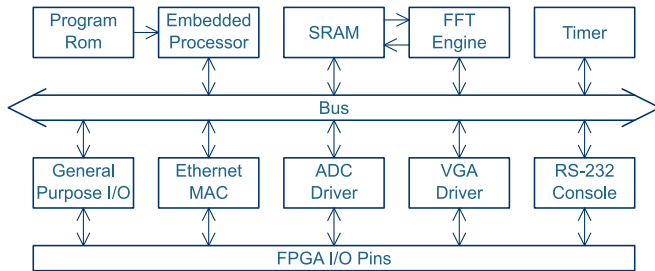
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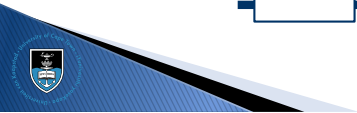
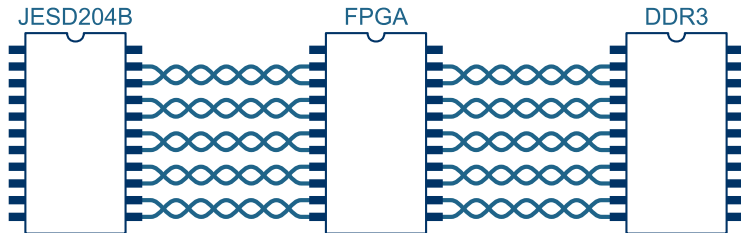
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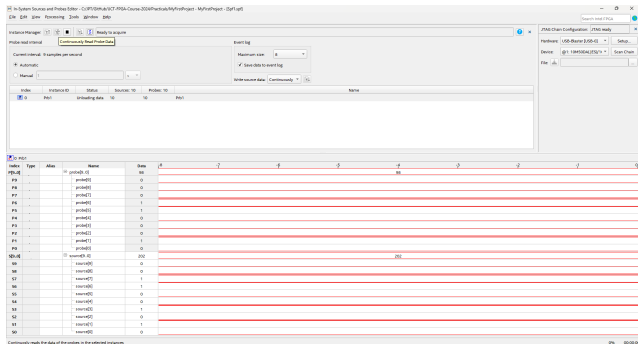
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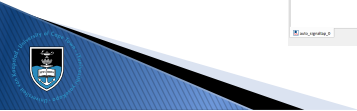


Debugging Infrastructure

- Sources and Probes (Practical 02 – Sources and Probes)
- Signaltap Logic Analyser (Practical 05 – SDRAM)
- Virtual JTAG (Practical 07 – PWM and Injection)

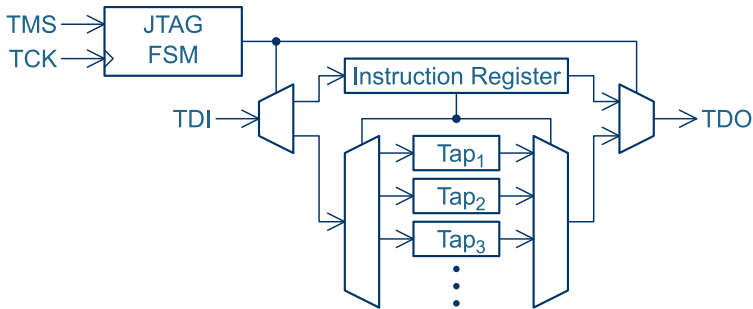


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Debugging Infrastructure

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- ▶ Virtual JTAG (Practical 07 – PWM and Injection)



Select References



Stephen Brown and Zvonko Vranesic
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ISBN 978-0-07-721164-6



Merrill L Skolnik
Introduction to RADAR Systems
ISBN 978-0-07-288138-7



Mark A. Richards and James A. Scheer
Principles of Modern Radar: Basic Principles
ISBN 978-1-89-112152-4



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World of ASIC
<http://www.asic-world.com/>



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FPGA 4 Fun
<http://www.fpga4fun.com/>



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Presented by Dr John-Philip Taylor
Convened by Dr Stephen Paine

Day 1 – 3 August 2026